REPORT

Lab-Assignment 1

Computer Networks

[ENCS352]

Submitted by:

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Course: BTech CSE – C

Submitted to:

Mr. Aijaz Mohammad Sir

Objective

To design and analyze different LAN topologies (Star, Hub-based, and Ring-like) using Cisco Packet Tracer, configure IPv4 addressing, test connectivity using ICMP, and study performance and fault tolerance behavior.

Learning Outcomes

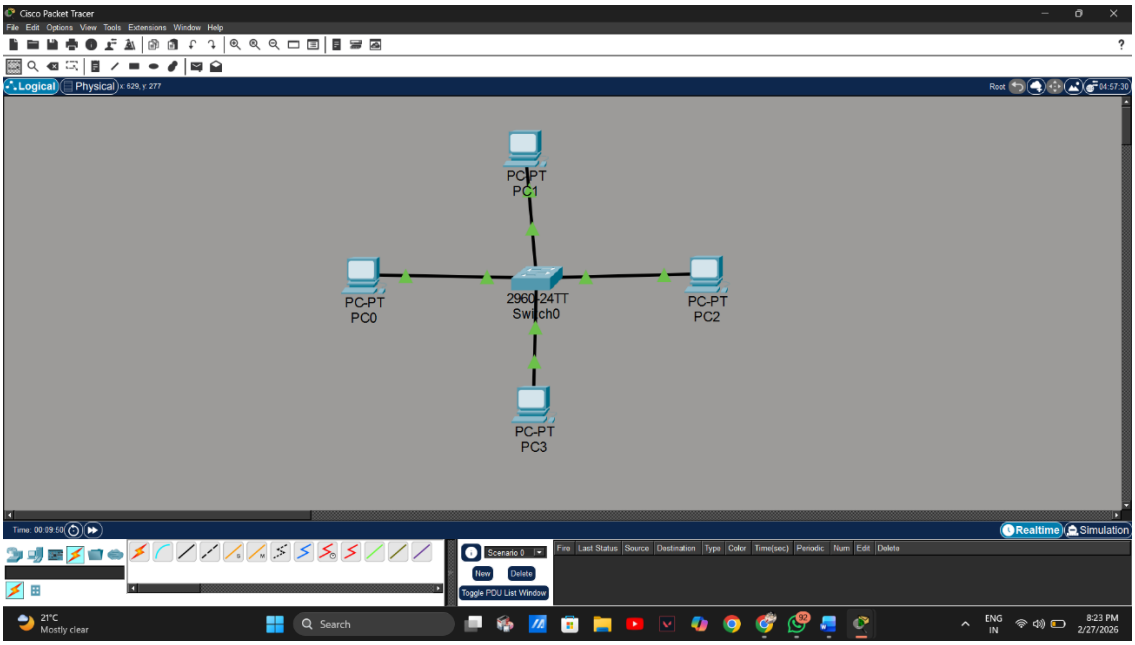
- Create small LAN networks in Packet Tracer
- Assign IPv4 addresses correctly
- Verify communication using ICMP (ping)
- Analyze collision domains and broadcast behavior
- Understand failure handling in different topologies

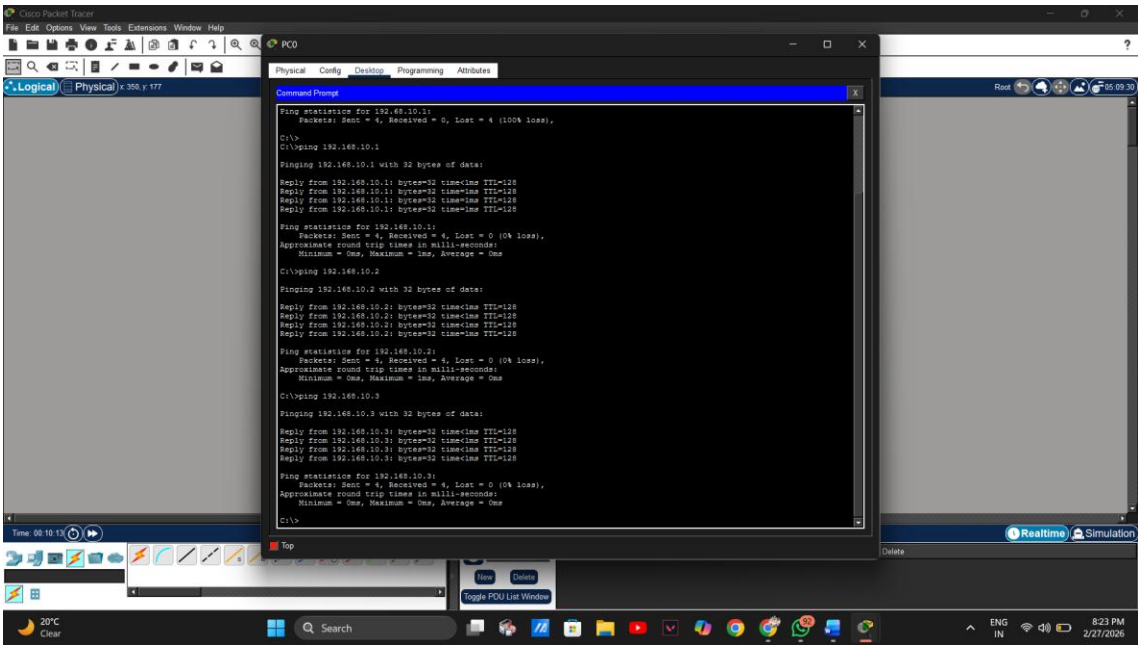
Concepts Used

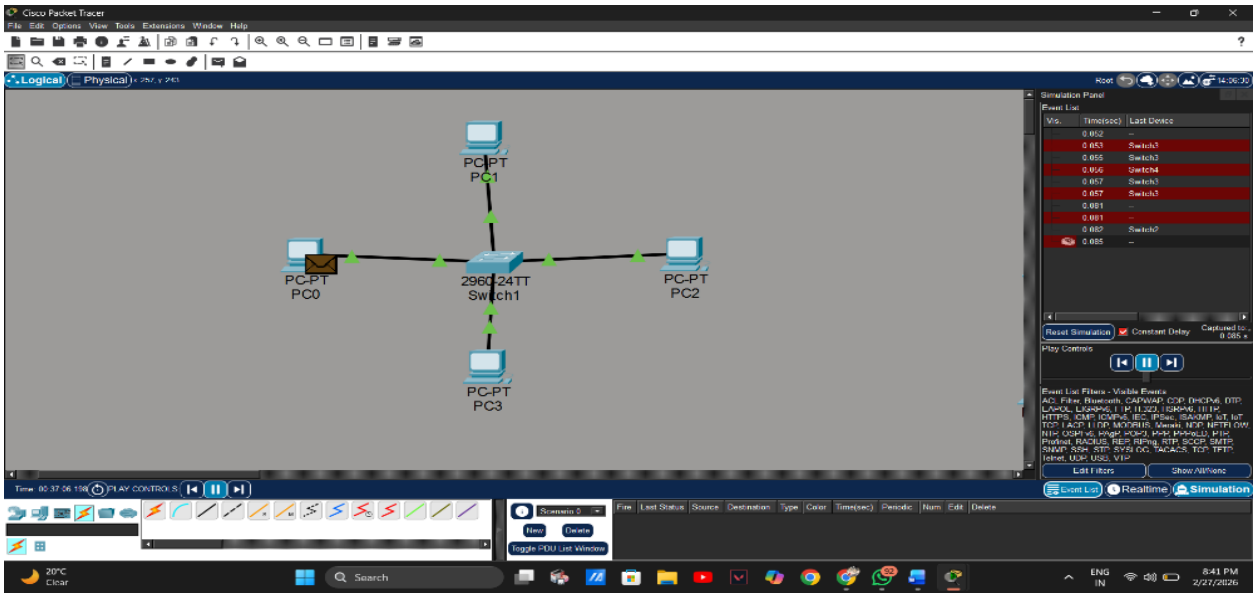
- Star Topology
- Bus-like Topology (Hub-based)
- Ring-like Topology (Switch Loop)
- IPv4 Addressing (Class C /24 network)
- ICMP Protocol
- Spanning Tree Protocol (STP)
- Packet Tracer Simulation Mode

Task 1: Star Topology (Switch)

- Place 1 switch + 4 PCs
- Connect all PCs to the switch
- Assign IPs (example): 192.168.10.1–4 /24
- Ping PC1 → PC2, PC3, PC4



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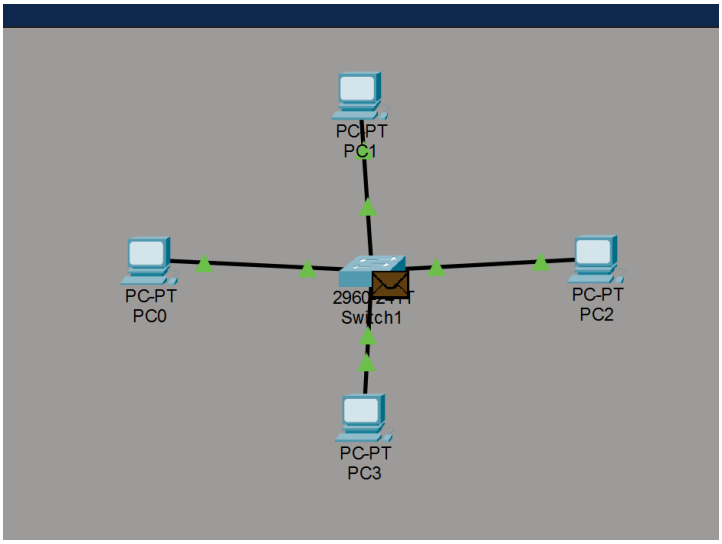
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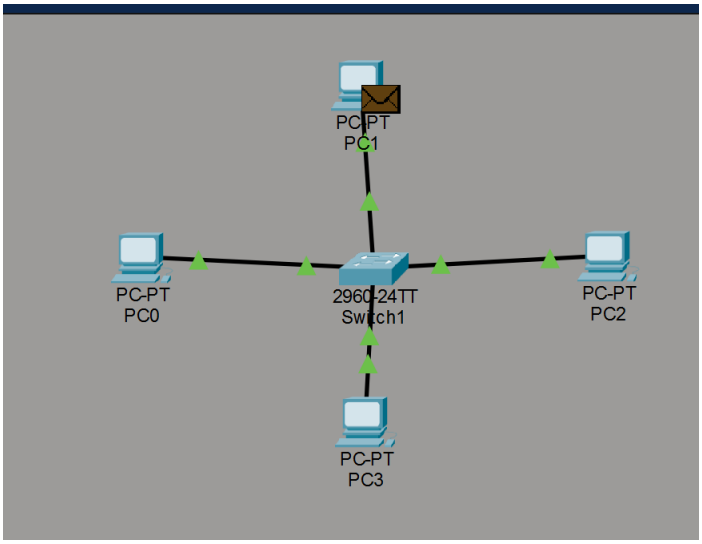
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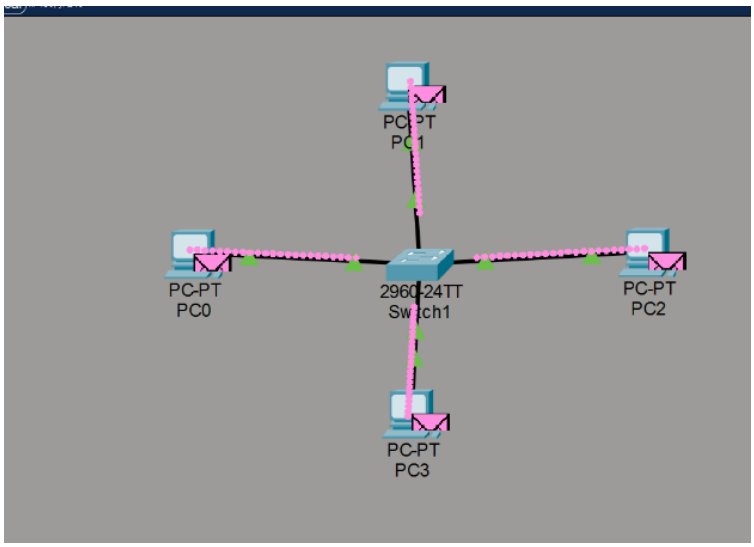
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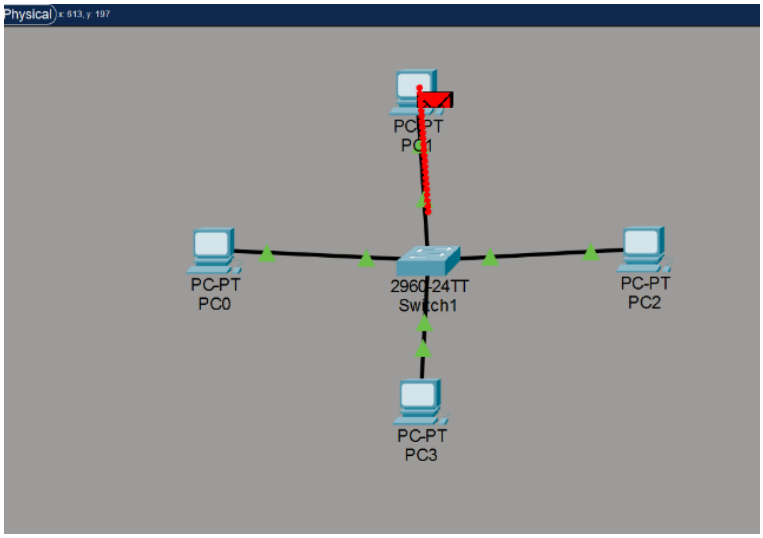
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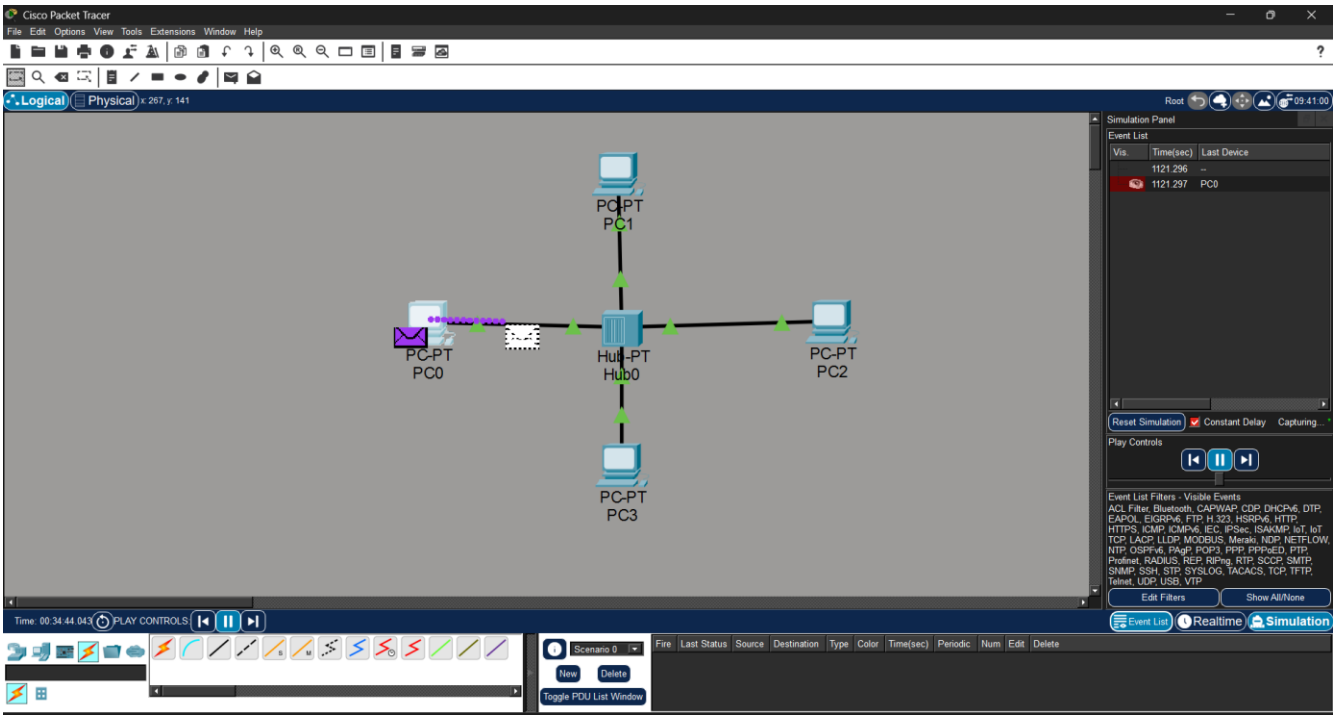
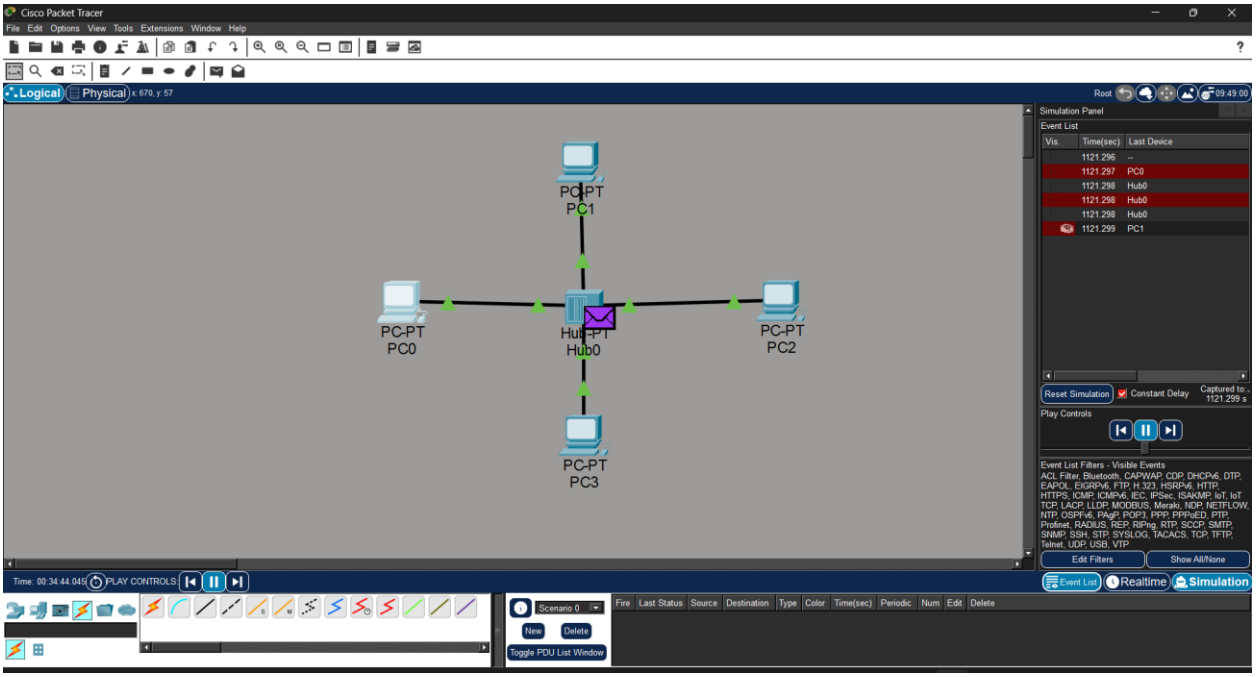
Observation

- All PCs successfully communicated.
- No packet loss observed.
- Switch forwards frames only to the destination port.
- Each port acts as a separate collision domain.
- If one PC cable disconnects → Only that PC is affected.
- If switch fails → Entire network fails.

Task 2: Bus-like Topology (Hub)

- Replace switch with hub
- Keep same IP scheme
- Send pings simultaneously and observe packet flow

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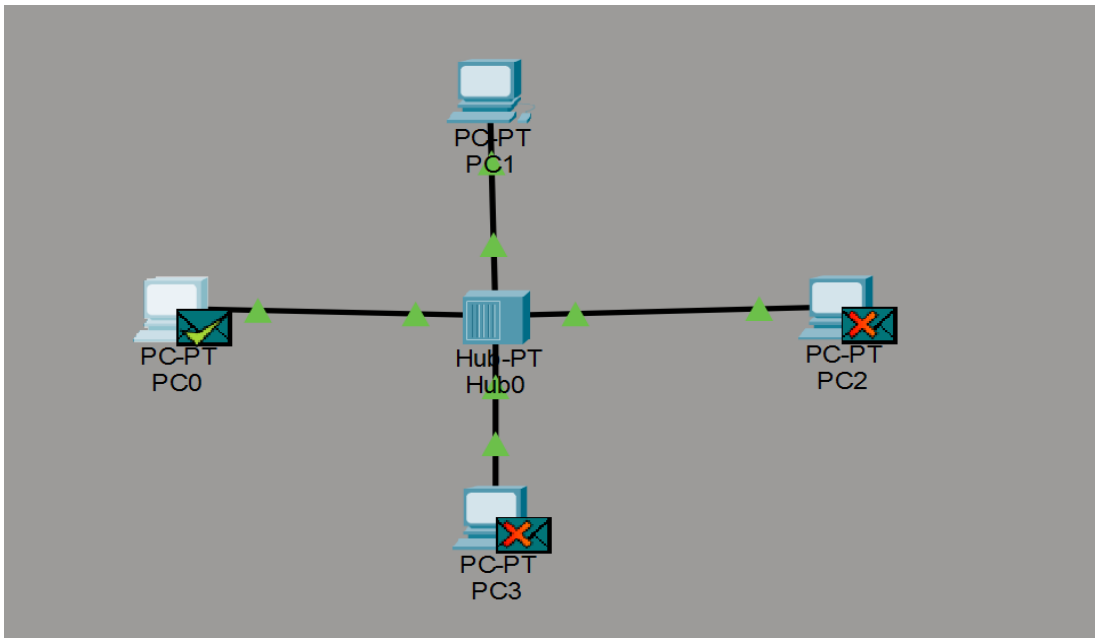
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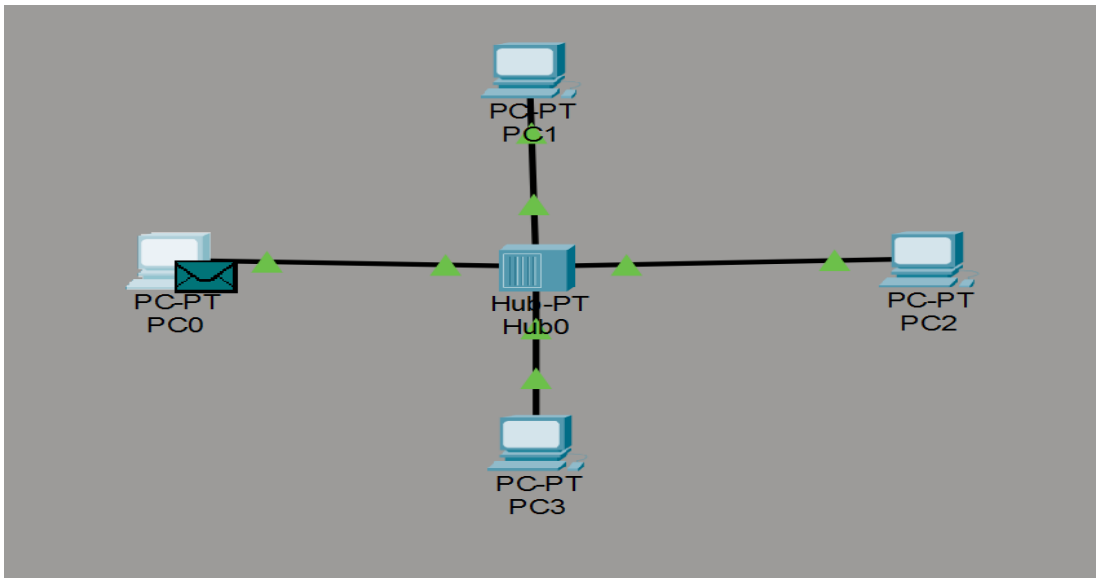
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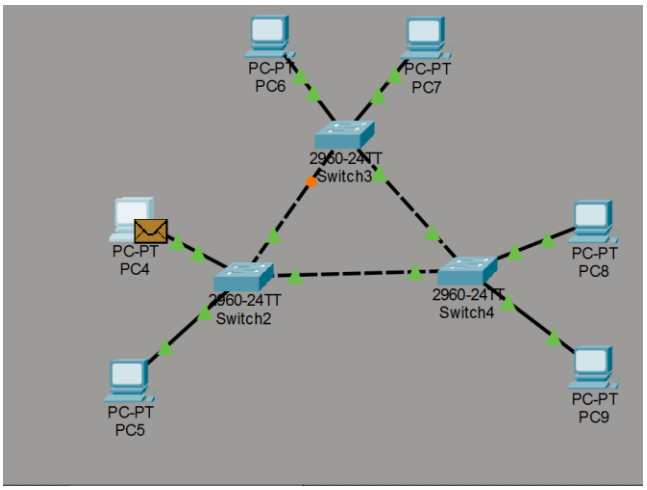
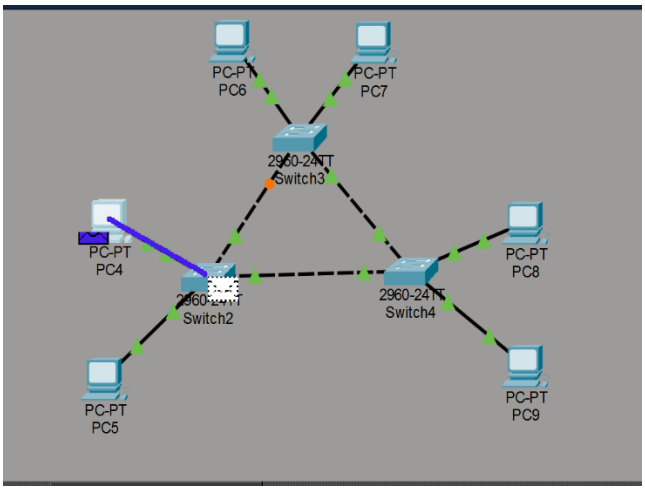
Observation

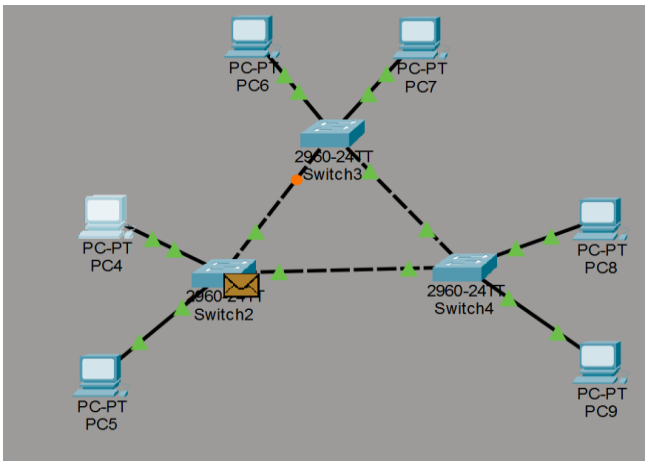
- All devices received broadcast packets.
- Hub flooded frames to all ports.
- Only intended device processed the packet.
- Single collision domain created.
- Shared bandwidth reduced performance.
- Hub failure results in complete network failure.

Task 3: Ring-like Topology (Loop)

- Place 3 switches and connect them in a loop
- Attach 1–2 PCs to each switch
- Ping across switches and observe path



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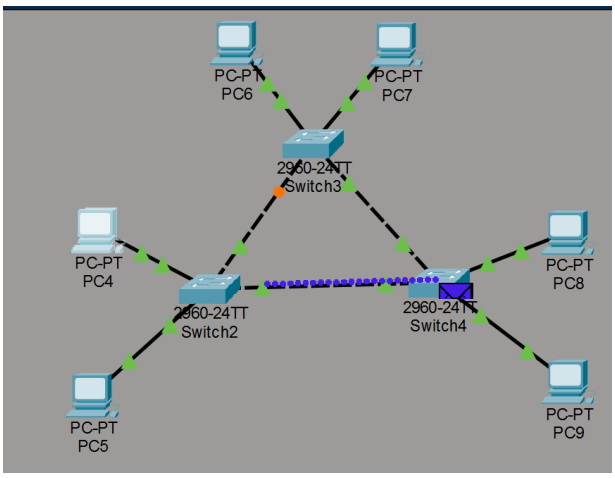
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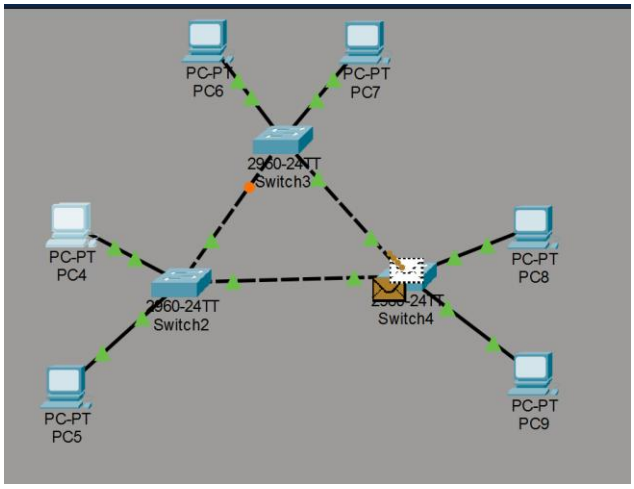
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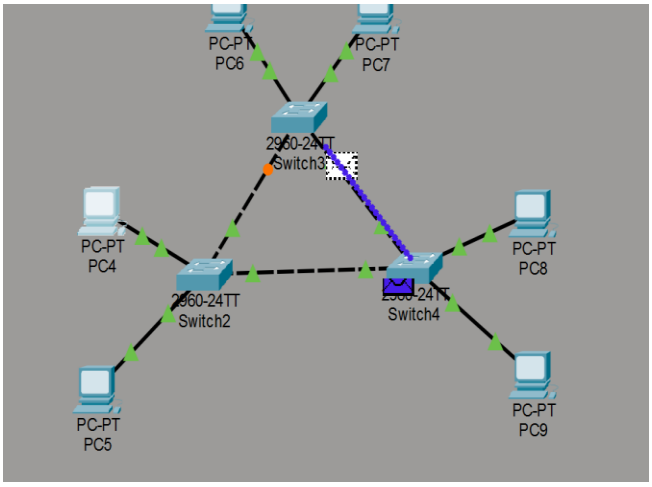
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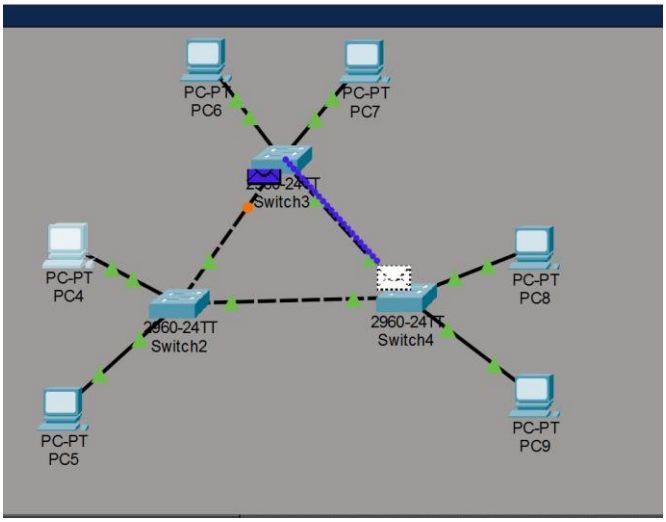
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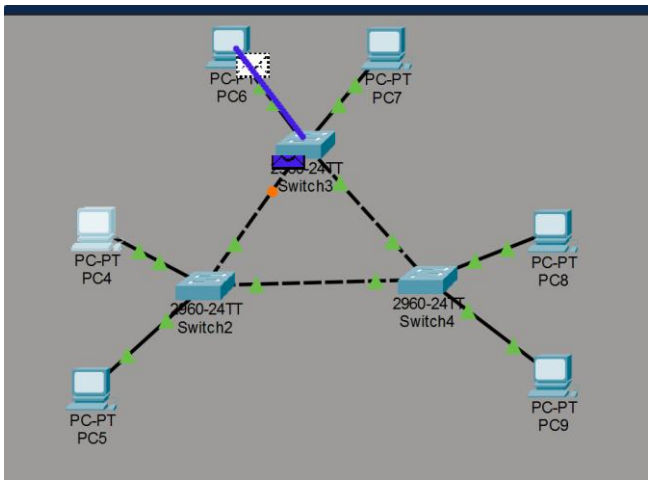
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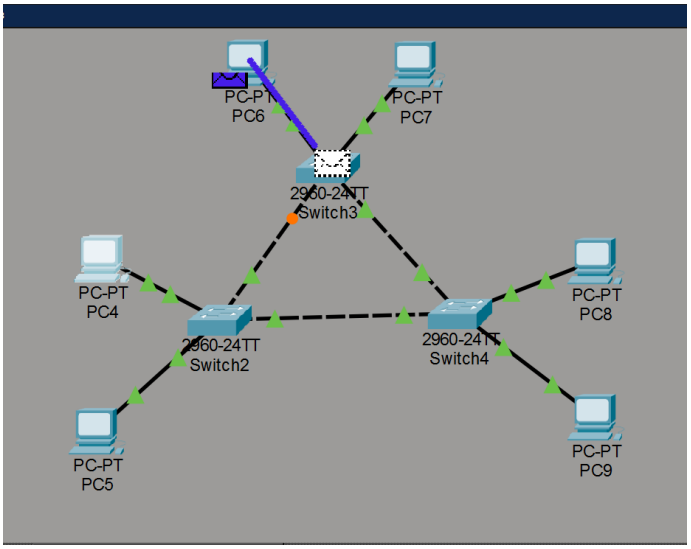
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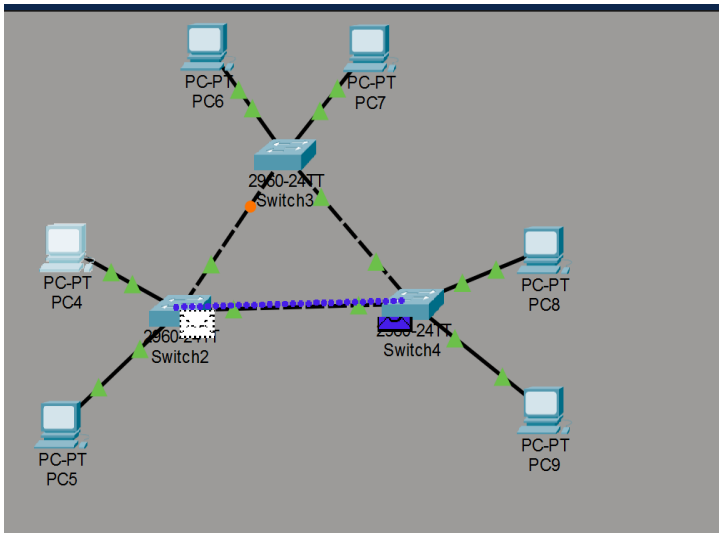
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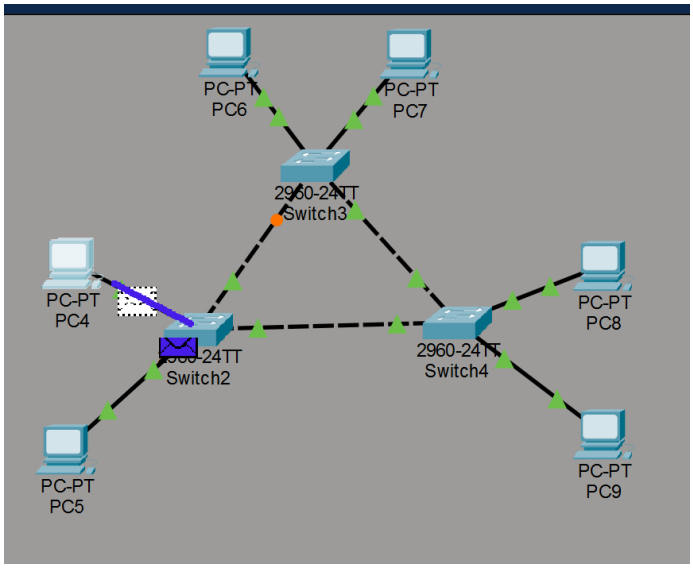
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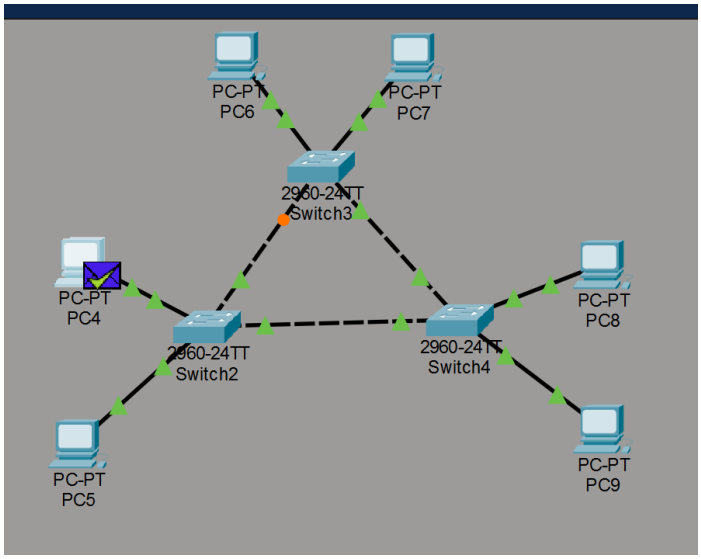
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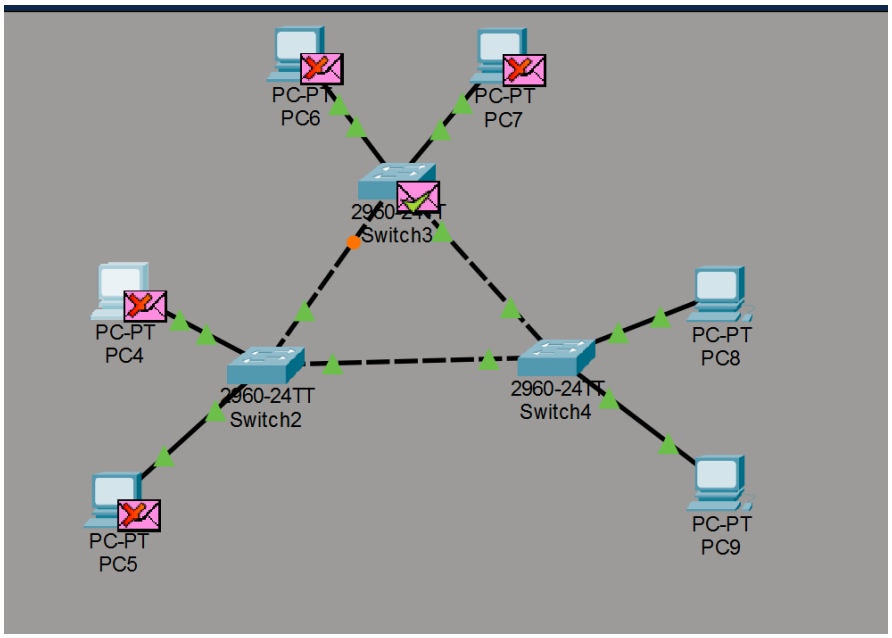
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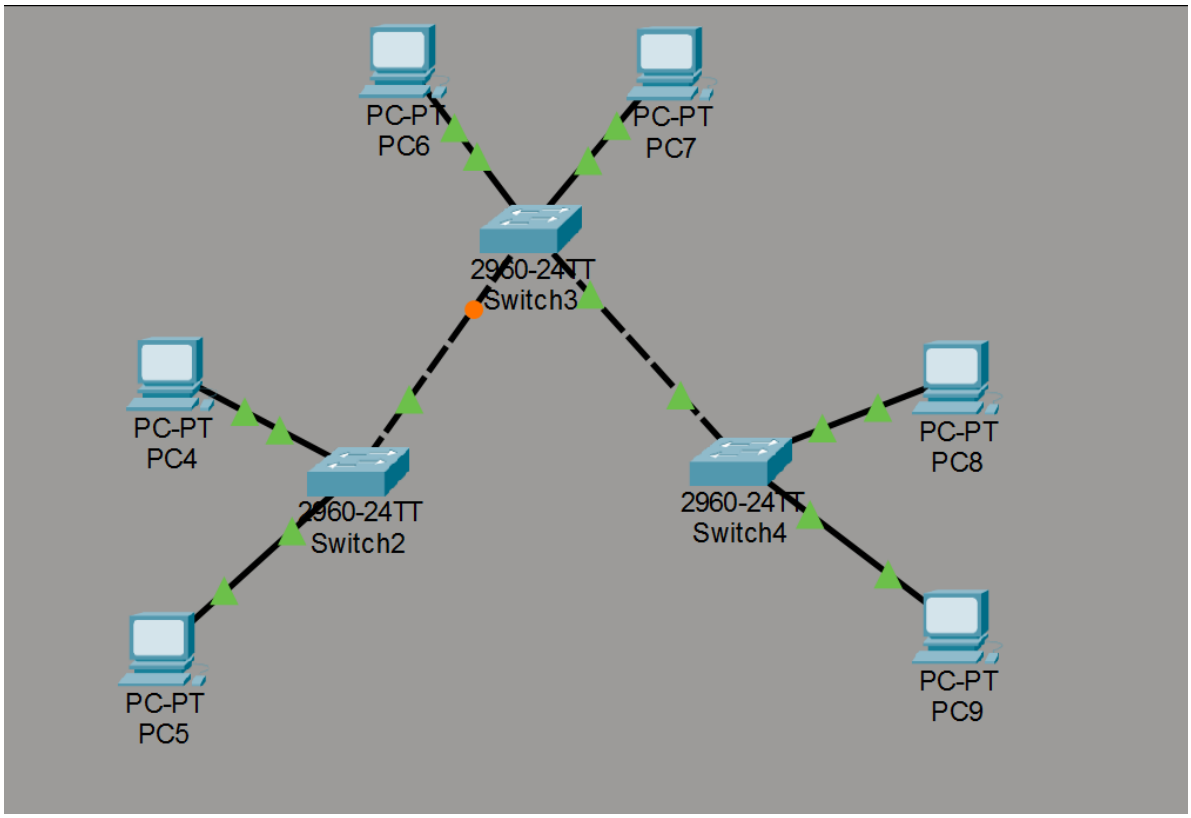
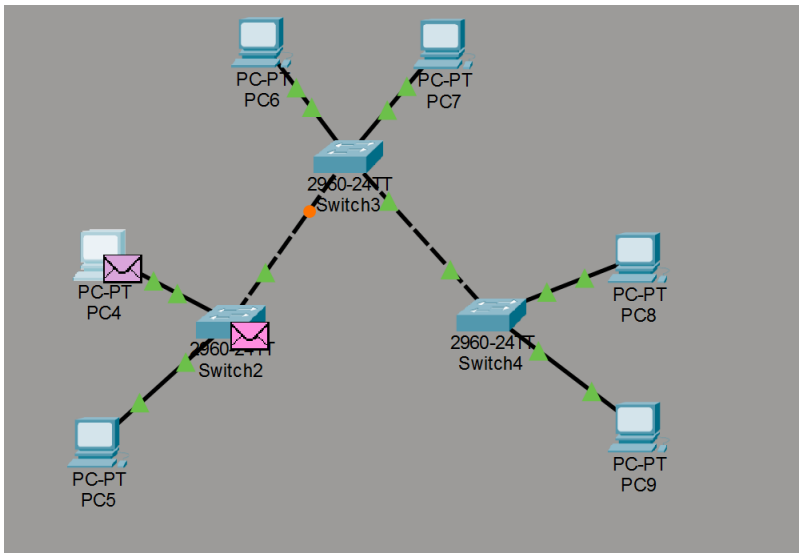
Observation

- Successful communication across switches.
- STP blocked one redundant port.
- Only one logical path remained active.
- Provides redundancy.
- Better fault tolerance than star and hub.

Task 4: Failure Test

- Disconnect one link in ring-like design
- Test if communication still works and note observations

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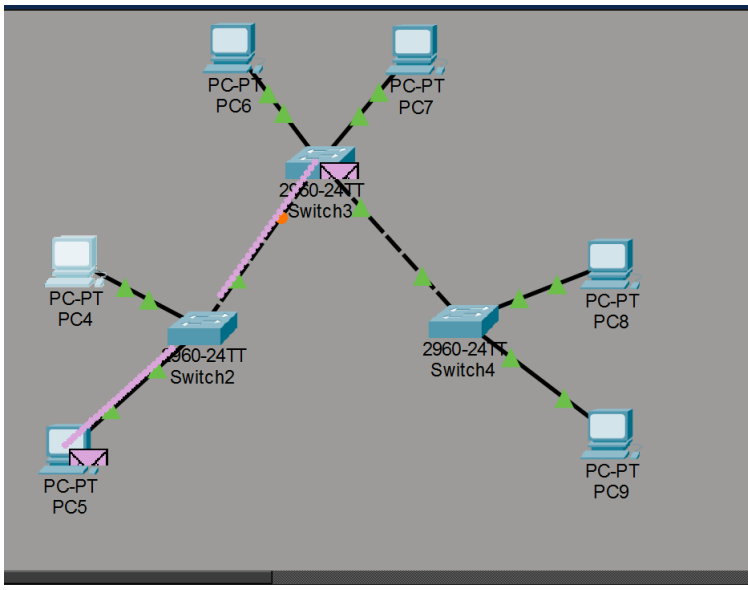
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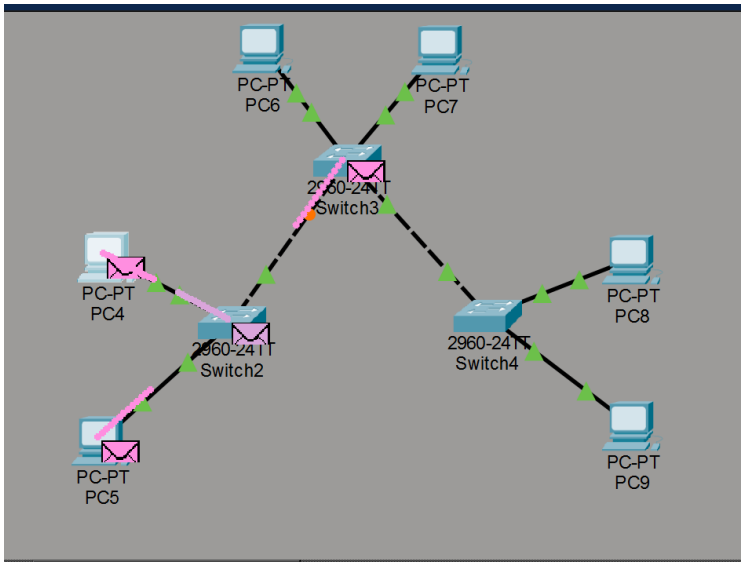
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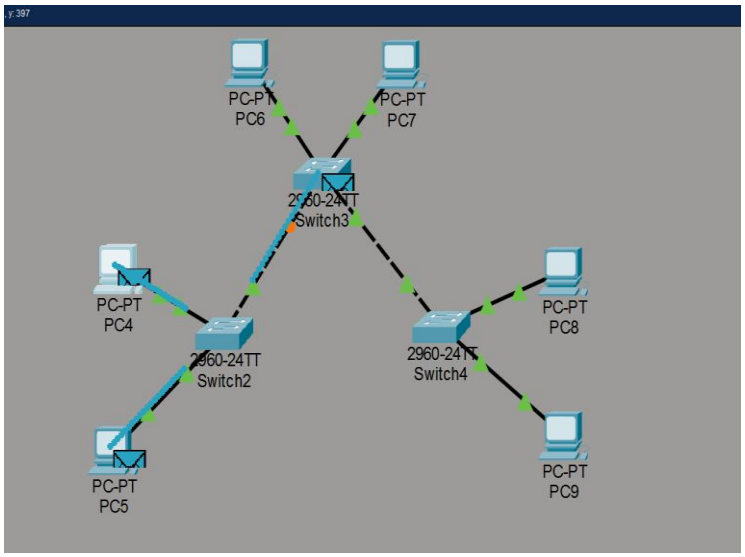
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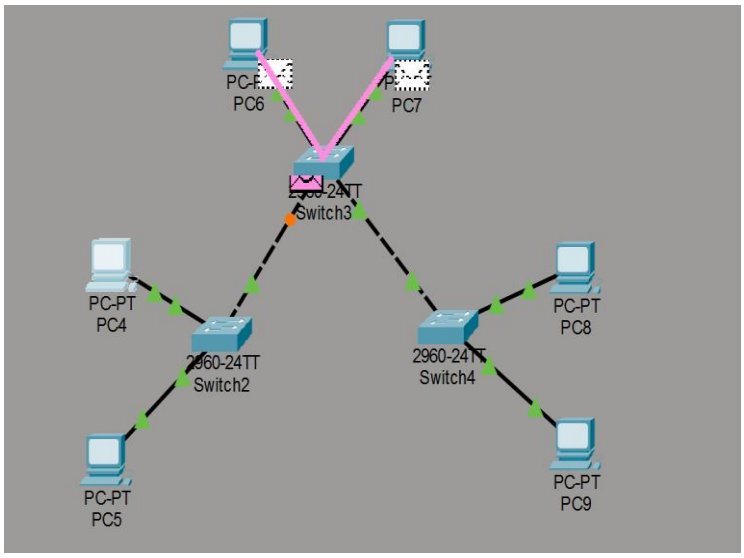
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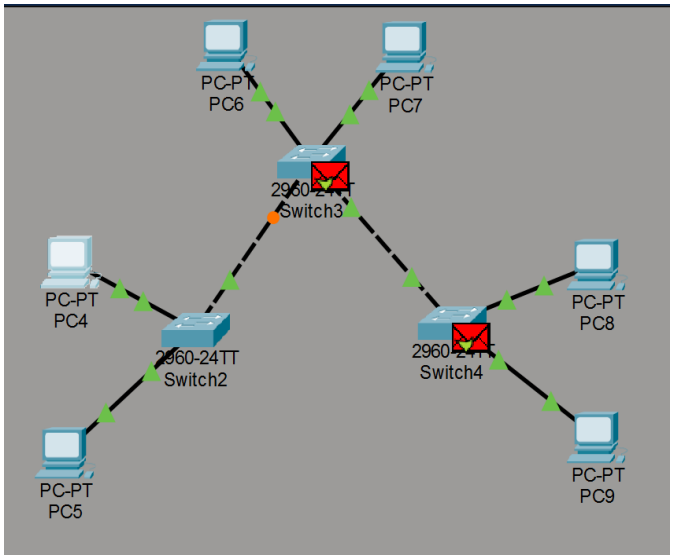
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Observation

- Communication continued successfully.
- STP recalculated topology.
- Previously blocked port became active.
- Network rerouted traffic automatically.
- Demonstrates fault tolerance and redundancy.

Conclusion

- This experiment demonstrated how network topology affects connectivity, performance, and reliability. Switch-based star topology offers better performance and easier troubleshooting. Hub-based topology creates broadcast traffic and lower efficiency. Ring-like topology provides redundancy and improved fault tolerance through Spanning Tree Protocol.
- Proper topology selection is essential for efficient and reliable LAN design.